MILITARY SPECIFICATION

LENS AND COVER; BEACON LIGHT, DRUM TYPE, 24 INCH

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers a 24-inch, drum type, white double-beam, beacon light lens and a 24-inch, drum-type rotating beacon cover.

2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specifications and standards. Unless otherwise specified (see 6.2), the following specifications and standards of the issue listed in that issue of the Department of Defense Index of Specifications and Standards (DoDISS) specified in the solicitation, form a part of this specification to the extent specified herein.

SPECIFICATIONS

FEDERAL

UU-P-268	Paper, Kraft, Wrapping
PPP-B-601	Boxes, Wood, Cleated-Plywood
PPP-B-636	Box, Shipping, Fiberboard

MILITARY

MIL-P-116	Preservation, Methods of
DOD-D-1000	Drawing, Engineering and Associated List
MIL-L-7158	Light, Beacon, Rotating, 24-Inch
MIL-C-7989	Covers, Light-Transmitting, For Aeronautical Lights, General Specification for
MIL-C-25050	Color, Aeronautical Lights And Lighting Equipment, General Requirements for

STANDARDS

MILITARY

MIL-STD-129	Marking for Shipment and Storage

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: the Engineering Division, San Antonio ALC/MMEDO, Kelly AFB, Texas 78241 by using the self addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

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MIL-STD-130	Identification Marking Of Property
MIL-STD-831	Test Reports, Preparation of
MS25015	Lamp, Incandescent, T-20 Bulb, Mogul Bipost Base
MS25391	Lens - 24 Inch Double Beam

AIR FORCE-NAVY AERONAUTICAL

AN2578	Cover, Drum-Type Rotating Beacon, 24-Inch

* (copies of specifications and standards required by manufacturers in connection with specific acquisition functions should be obtained from the contracting activity or as directed by the contracting officer.)

* 2.1.2 Other publications. The following document(s) form a part of this specification to the extent specified herein. The issues of the documents which are indicated as DoD adopted shall be the issue listed in the current DoDISS and the supplement thereto, if applicable.

* AMERICAN SOCIETY FOR TESTING AND MATERIALS (ASTM)

* ASTM D 3951 Standard Practice for Commercial Packaging

* (Application for copies of ASTM publications should be addressed to the American Society for Testing and Materials, 1916 Race Street, Philadelphia, Pennsylvania 19103.)

* 2.1.3 Order of precedence. In the event of a conflict between the text of this specification and the references cited herein, the text of this specification shall take precedence.

3. REQUIREMENTS

3.1 Preproduction testing. This specification makes provisions for preproduction testing.

3.2 Materials. Material shall conform to the applicable specifications as specified herein. Materials which are not specifically designated shall be of the best commercial quality and suitable for the purpose intended.

3.2.1 Glass. Glass shall be in accordance with the requirements for class B of MIL-C-7989.

3.2.2 Color. Color shall be in accordance with MS25391 and AN2578 (see 6.2) and shall comply with the requirements for type I(d) of MIL-C-25050 for the applicable AN part number.

* 3.2.3 Reclaimed Materials. Reclaimed materials shall be used to the maximum extent possible without jeopardizing the intended use of the item.

3.3 Design and construction.

3.3.1 Double beam white lens. The dimensions of the lens shall be in accordance with MS25391. When used with a 1,200 watt (w), 27,500 lumina, MS25015-2 lamp and a 24-inch beacon which complies with the photometric and

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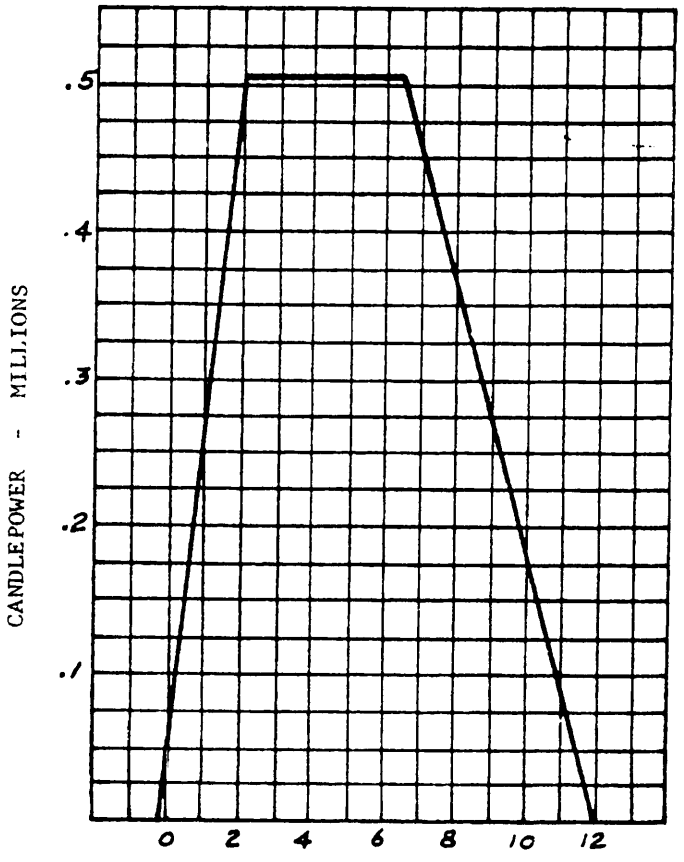


FIGURE 1 -LIGHT DISTRIBUTION

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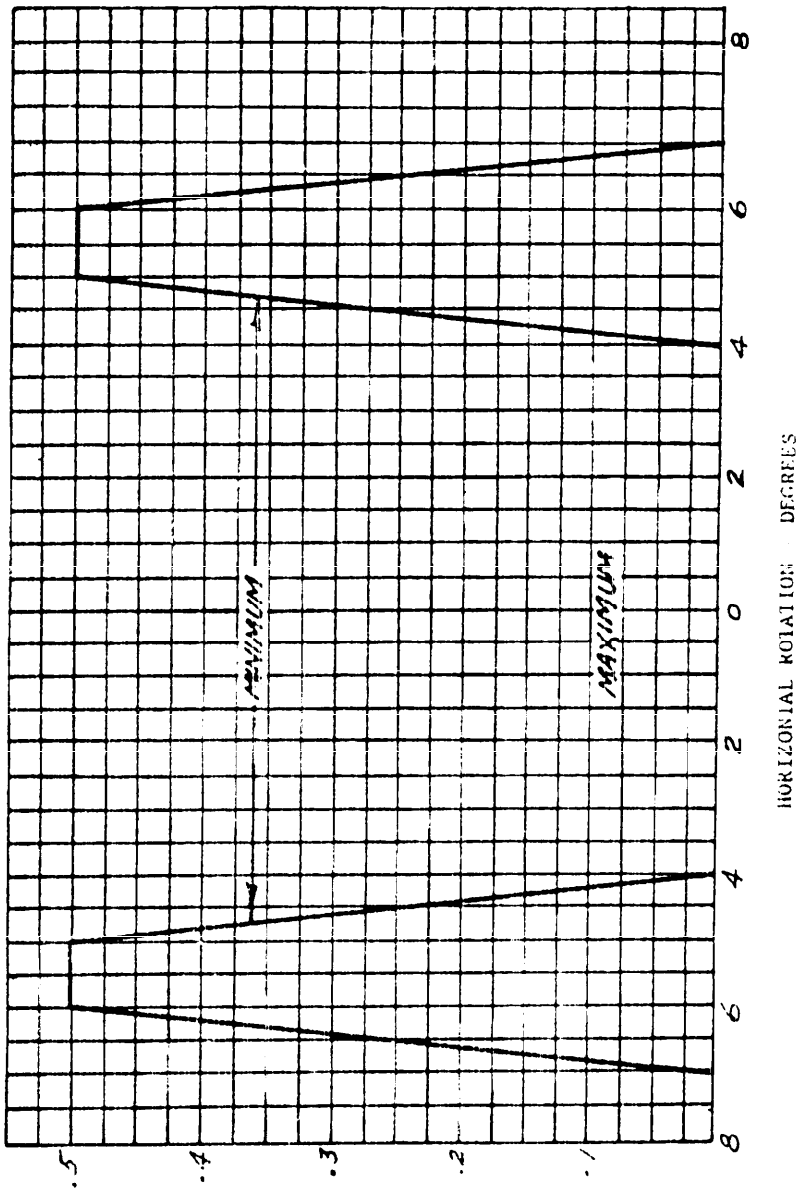


FIGURE 2 - LIGHT DISTRIBUTION

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optical requirements of MIL-L-7158, the lens shall produce a light distribution which equals or exceeds the minimum values and does not exceed the maximum values, as shown on Figures 1 and 2. The peaks of the two beams shall be separated 11 degrees ±2 degrees horizontally, and shall be not less than 700,000 candlepower. The beam width at the 10 percent zone shall not exceed 6 degrees for horizontal rotation.

3.3.2 Single-beam covers. The dimensions of the covers shall be in accordance with AN2578. When used with a 1,200w, 115 volts, 27,500 lumina, MS25015-2 lamp and a 24-inch beacon which complies with the photometric and optical requirements of MIL-L-7158, the covers shall have an average transmission ratio (measured at a minimum of 10 different points after the covers have reached a stable temperature and when the assembly is operated in an ambient temperature of approximately 20° C) to equal or exceed the minimum acceptable transmission ratio for type I(d) glassware as specified in MIL-C-25050 for the specified color.

3.4 Interchangeability. All parts having the same manufacturer's part number shall be functionally and dimensionally interchangeable. The drawing number requirements of DOD-D-1000 shall govern changes in the manufacturer's part number.

3.5 Markings. The following markings, as applicable, shall be permanently molded into the lens or cover in such a position that they can be read when the lens or cover is installed:

24"	24"
Double Beam	or AN2578-
MS25391-1	(add dash number)

3.6 Identification of product. Equipment, assemblies, and parts shall be marked for identification in accordance with MIL-STD-130.

3.7 Workmanship. Workmanship shall be in accordance with high-grade commercial practice covering this class of work.

4. QUALITY ASSURANCE PROVISIONS

* 4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.2 Classification of tests. The inspection and testing of lens and covers shall be classified as follows:

a. Preproduction tests.

b. Quality conformance inspection.

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4.3 Preproduction testing.

4.3.1 Preproduction test sample tested by the contractor. The contractor shall subject one lens and cover to the preproduction tests specified in 4.3.4.

4.3.2 Test report. After the contractor completes the preproduction tests, he shall prepare a preproduction test report in accordance with MIL-STD-831 and furnish three complete copies of the report to the preparing activity.

4.3.3 Preproduction test sample for the procuring activity. Along with the preproduction test report, the contractor shall submit the sample to the procuring activity.

4.3.4 Preproduction tests. Preproduction tests shall consist of all tests described under 4.5 and as specified in MIL-C-7989 and MIL-C-25050.

4.4 Quality conformance inspection. The quality conformance inspection shall consist of the individual tests.

4.4.1 Individual tests. Each lens and cover shall be subjected to the following tests as described under 4.5:

a. Examination of product.

b. Photometric.

4.5 Test methods.

4.5.1 Examination of product. Each unit submitted for acceptance under contract shall be visually examined to determine conformance with this specification with respect to material and workmanship. This shall include a physical examination of the glassware and a determination as to whether the dimensions are correct.

4.5.2 Photometric test. Photometric tests shall be conducted by the contractor to determine compliance with the photometric requirements of this specification (see 3.3.1 and 3.3.2).

4.6 Inspection of the preservation, packaging, packing and marking for shipment and storage. Sample items or packs and the inspection of the preservation, packaging, packing and marking for shipment and storage shall be in accordance with the requirements of Section 5, or the documents specified therein.

5. PACKAGING

* 5.1 Preservation. Preservation shall be level A or commercial, as specified (see 6.2).

5.1.1 Level A. Lenses or covers shall be individually packaged Method III of MIL-P-116. Each lens or cover shall be wrapped with kraft wrapping paper conforming to UU-P-268 to prevent scoring of surfaces and shall be adequately cushioned, blocked, braced, et cetera, to prevent physical damage. Unit containers shall conform to PPP-B-636, class weather resistant. The unit container, packaged as for shipment, shall meet the rough handling tests of MIL-P-116.

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* 5.1.2 Commercial. Lenses or covers shall be preserved in accordance with ASTM D 3951.

5.2 Packing.. Packing shall be level A, B, or commercial as specified (see 6.2).

5.2.1 Level A. Quantities of like items packaged as specified in 5.1.1 shall be packed in wood cleated-plywood containers conforming to PPP-B-601, overseas type. As far as practicable, exterior containers shall be uniform in shape and size, of minimum cube and tare consistent with the protection required, and contain identical quantities. The gross weight of each shipping container when packed for shipment shall not exceed approximately two hundred (200) pounds. Container closure and strapping shall be in accordance with the appendix to PPP-B-601.

5.2.2 Level B. Lenses or covers packaged to meet requirements of 5.1.1 shall require no further overpacking.

* 5.2.3 Commercial. Lenses or covers preserved as specified in 5.1, shall be packed in accordance with ASTM D 3951.

* 5.3 Marking. In addition to any special marking required by the contract or order, interior containers and exterior shipping containers shall be marked in accordance with MIL-STD-129 or ASTM D 3951 as applicable. The following marking not covered by MIL-STD-129 or ASTM D 3951 as applicable shall also be added:

AN **

**Add applicable AN standard and dash number.

6. NOTES

6.1 Intended use. The units covered by this specification are intended for use on 24-inch rotating airport beacon lights which provide sufficient vertical coverage for high altitude jet aircraft. Their use for other purposes is prohibited.

* 6.2 Ordering data.

* 6.2.1 Acquisition requirements. Acquisition documents should specify the following:

a. AN part number of the lens and covers desired (see 3.2.2).

b. Selection of applicable levels of packaging and packing (see 5.1 and 5.2).

* 6.3 Changes from previous issue. The margins of this specification are marked with asterisks to indicate where changes (additions, modifications, corrections, deletions) from the previous issue were made. This was done as a convenience only and the Government assumes no liability whatsoever for any inaccuracies in these notations. Bidders and contractors are cautioned to evaluate the requirements of this document based on the entire content irrespective of the marginal notations and relationship to the last previous issue.

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5. PROBLEM AREAS

a. Paragraph Number and Wording:

b. Recommended Wording:

c. Reason/Rationale for Recommendation:

6. REMARKS

7a. NAME OF SUBMITTER *(Last, First, MI)* – Optional	b. WORK TELEPHONE NUMBER *(Include Area Code)* – Optional
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